# STT180 Honors Project

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## NCAA Success and Expenses

```
library(tidyverse)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

### Data Cleaning

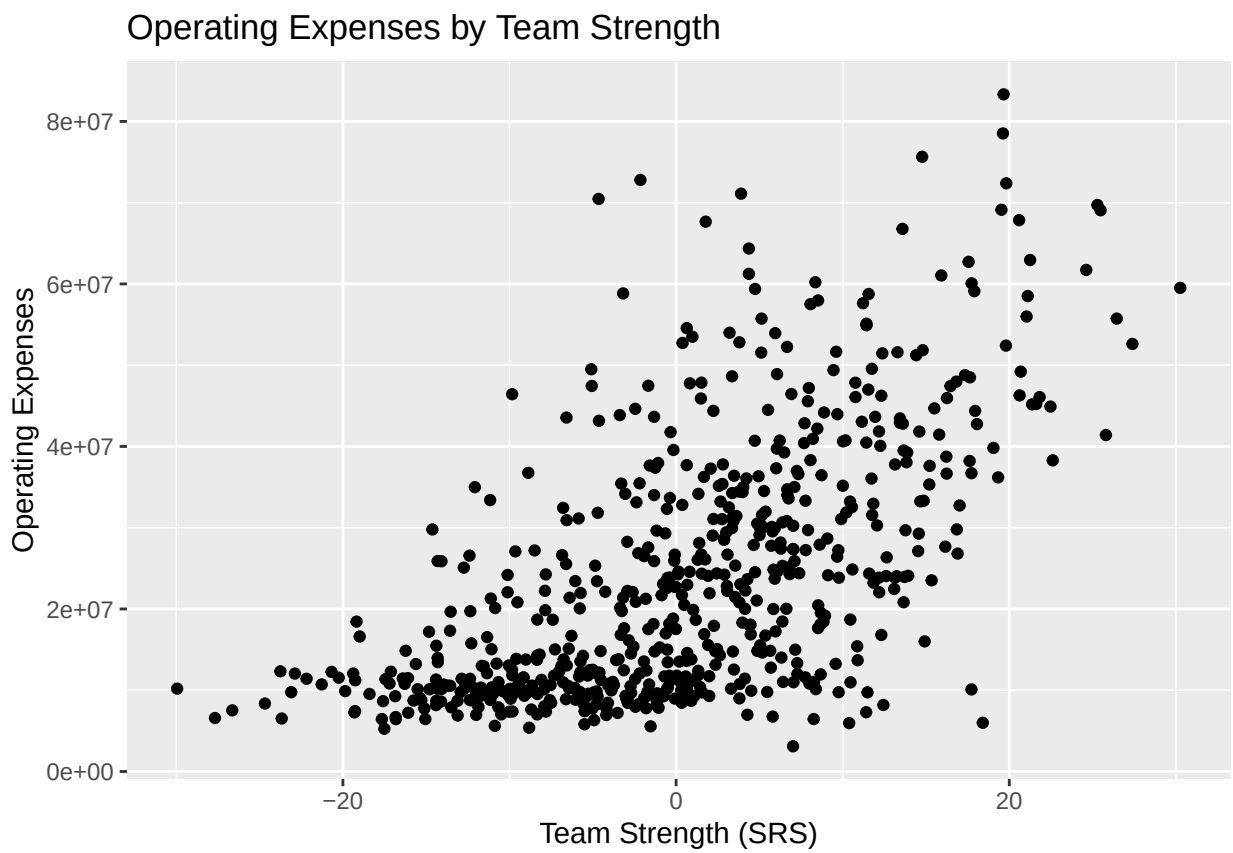
```
CFBdata <- read.csv("NCAARevenueData.csv")
CFBdata$Football.Total.Operating.Expenses <- as.numeric(gsub('[$,]', '', CFBdata$Football.Total.Operati
```

### Data Visualization

#### Team Strength vs Operating Expenses

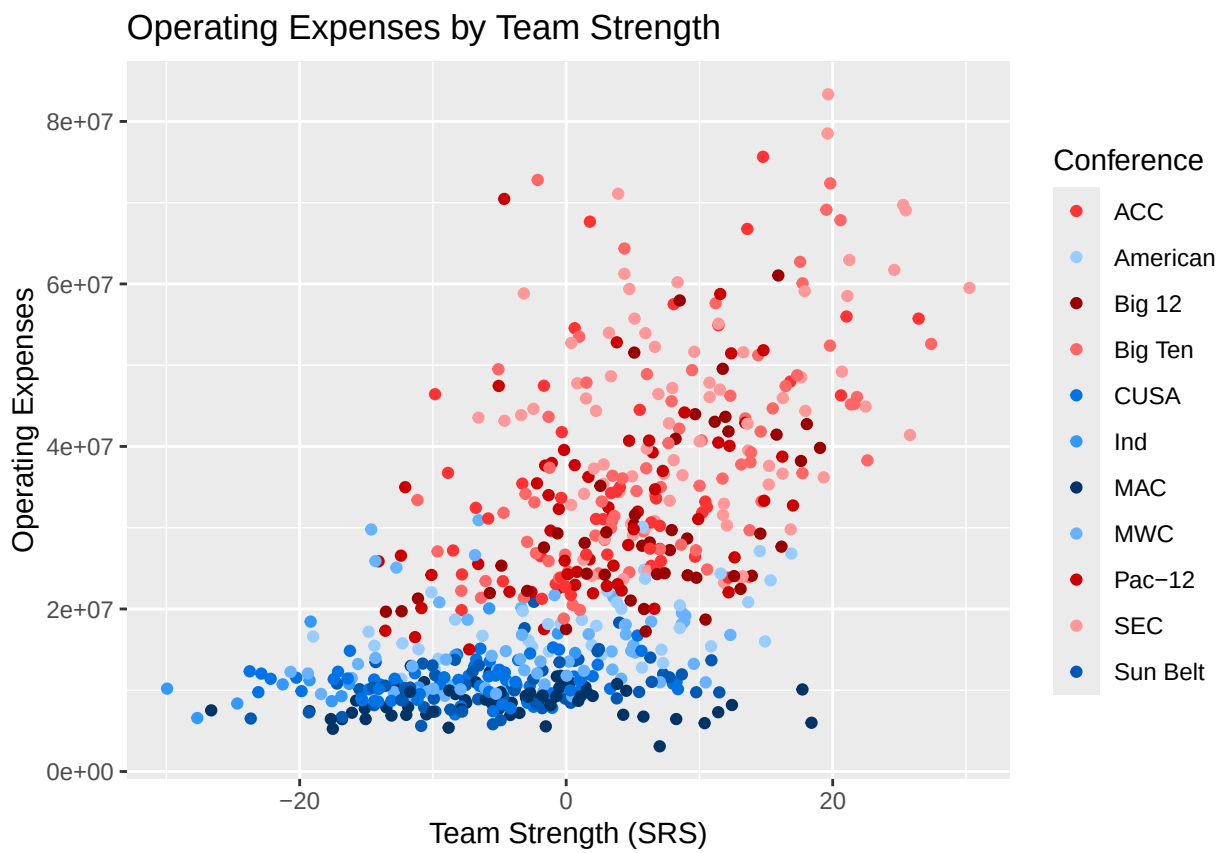
```
ggplot(CFBdata, aes(SRS, Football.Total.Operating.Expenses)) +
  geom_point() +
  labs(title = "Operating Expenses by Team Strength", x = "Team Strength (SRS)", y = "Operating Expense
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



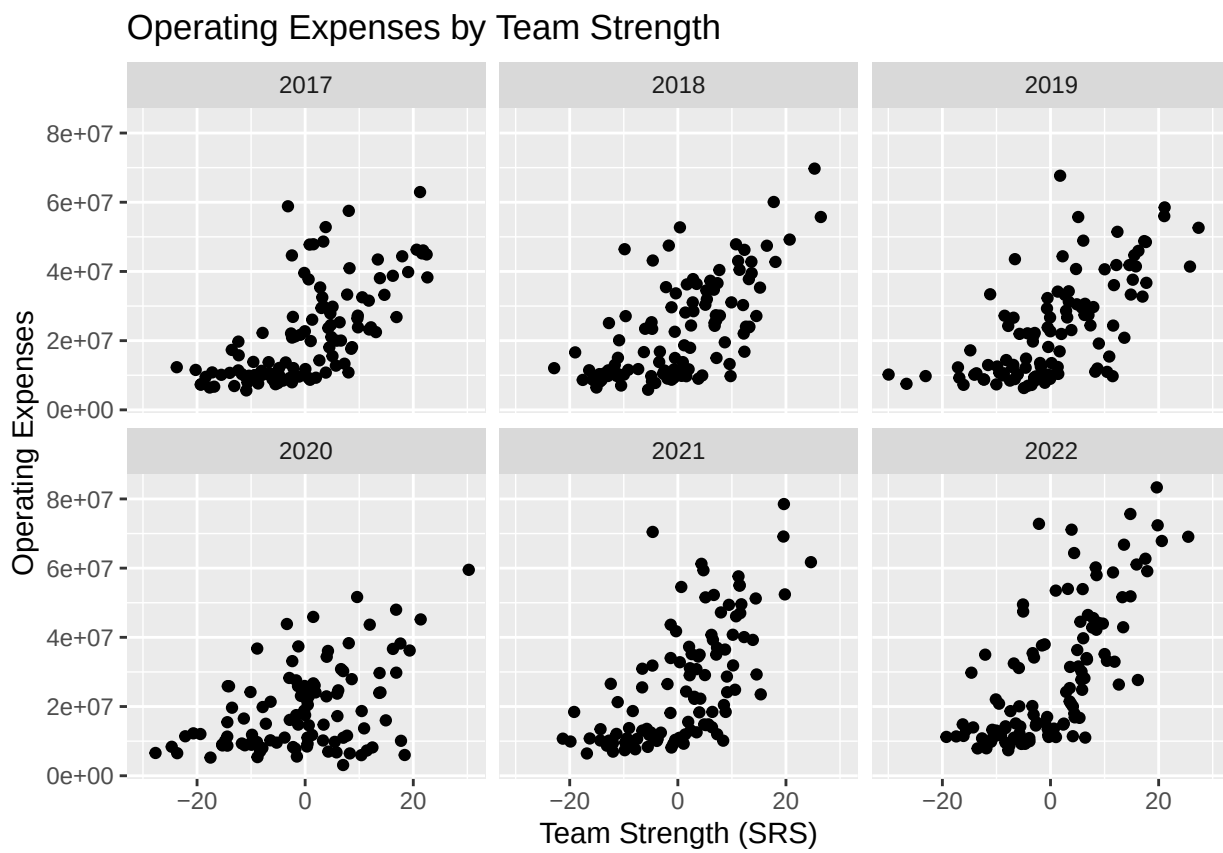

```
ggplot(CFBdata, aes(SRS, Football.Total.Operating.Expenses, color=factor(Conf))) + geom_point() +
  scale_color_manual(values = c(
    "SEC" = "#FF9999",
    "Big Ten" = "#FF6666",
    "ACC" = "#FF3333",
    "Pac-12" = "#CC0000",
    "Big 12" = "#990000",
    "American" = "#99CCFF",
    "MWC" = "#66B2FF",
    "Ind" = "#3399FF",
    "CUSA" = "#0073E6",
    "Sun Belt" = "#0059B3",
    "MAC" = "#003366"
  )) +
  labs(title = "Operating Expenses by Team Strength", x = "Team Strength (SRS)", y = "Operating Expense
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggplot(CFBdata, aes(SRS, Football.Total.Operating.Expenses)) + geom_point() + facet_wrap(~ Year) +
  labs(title = "Operating Expenses by Team Strength", x = "Team Strength (SRS)", y = "Operating Expense
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




**Slope Fit / By-Year**

```
mod.fit17 <- lm(Football.Total.Operating.Expenses~SRS, data=subset(CFBdata, Year == 2017))

mod.fit18 <- lm(Football.Total.Operating.Expenses~SRS, data=subset(CFBdata, Year == 2018))

mod.fit19 <- lm(Football.Total.Operating.Expenses~SRS, data=subset(CFBdata, Year == 2019))

mod.fit20 <- lm(Football.Total.Operating.Expenses~SRS, data=subset(CFBdata, Year == 2020))

mod.fit21 <- lm(Football.Total.Operating.Expenses~SRS, data=subset(CFBdata, Year == 2021))

mod.fit22 <- lm(Football.Total.Operating.Expenses~SRS, data=subset(CFBdata, Year == 2022))
```

```
mod.fit17 %>%
  broom::glance()
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared     sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl>     <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.442         0.437 10642022.      83.2 5.78e-15     1 -1882. 3770. 3778.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
mod.fit18 %>%
  broom::glance()
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared     sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl>     <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.421         0.415 10949147.      75.5 5.67e-14     1 -1868. 3741. 3749.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
mod.fit19 %>%
  broom::glance()
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared     sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl>     <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.434         0.429 11035750.      80.6 1.19e-14     1 -1886. 3778. 3786.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
mod.fit20 %>%
  broom::glance()
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared     sigma statistic   p.value    df logLik   AIC   BIC
##       <dbl>         <dbl>     <dbl>     <dbl>     <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.209         0.202 10808488.      27.2   9.35e-7     1 -1849. 3703. 3711.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
mod.fit21 %>%
  broom::glance()
```

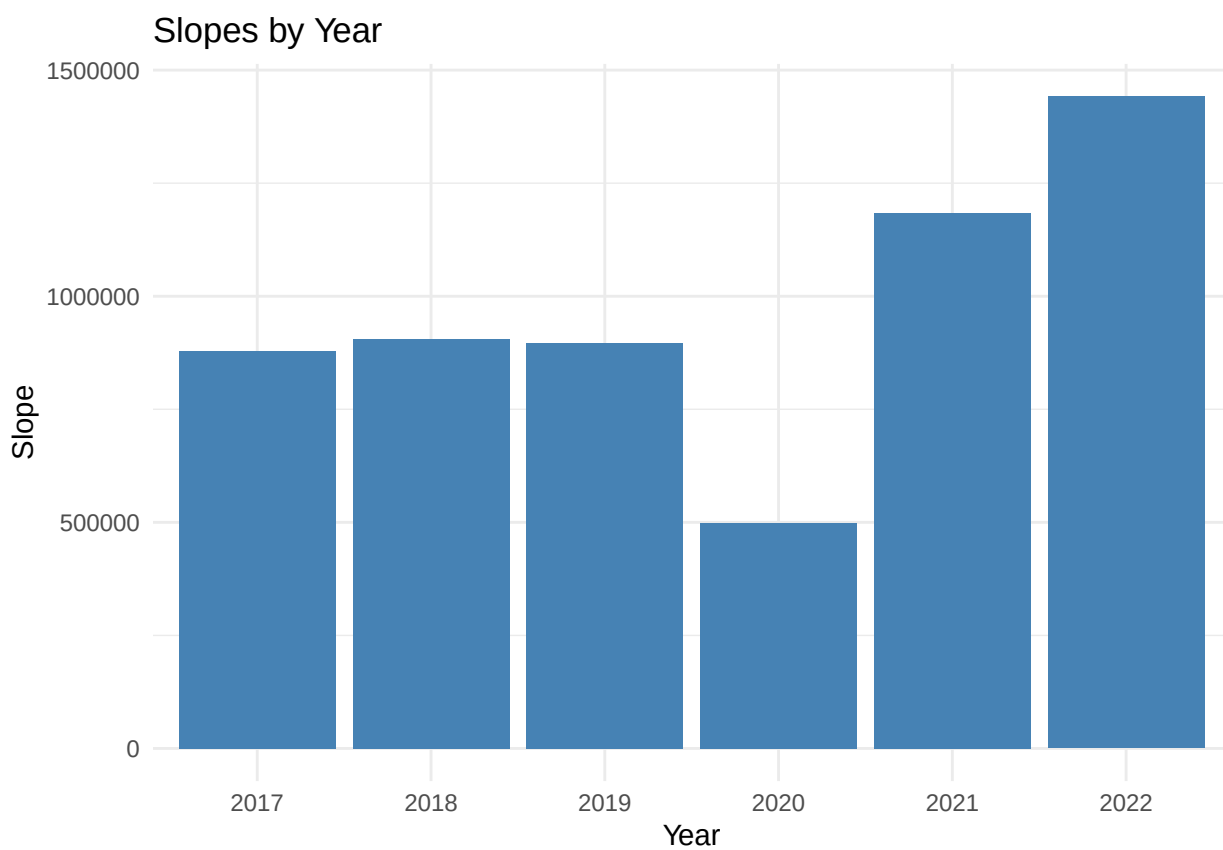
```
## # A tibble: 1 x 12
##   r.squared adj.r.squared     sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl>     <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.417         0.412 13339956.      74.5 7.51e-14     1 -1888. 3783. 3791.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
mod.fit22 %>%
  broom::glance()
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared     sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl>     <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.494         0.489 14054937.      105. 1.15e-17     1 -1966. 3937. 3945.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
library(broom)

slope17 <-  tidy(mod.fit17)$estimate[2]
slope18 <-  tidy(mod.fit18)$estimate[2]
```

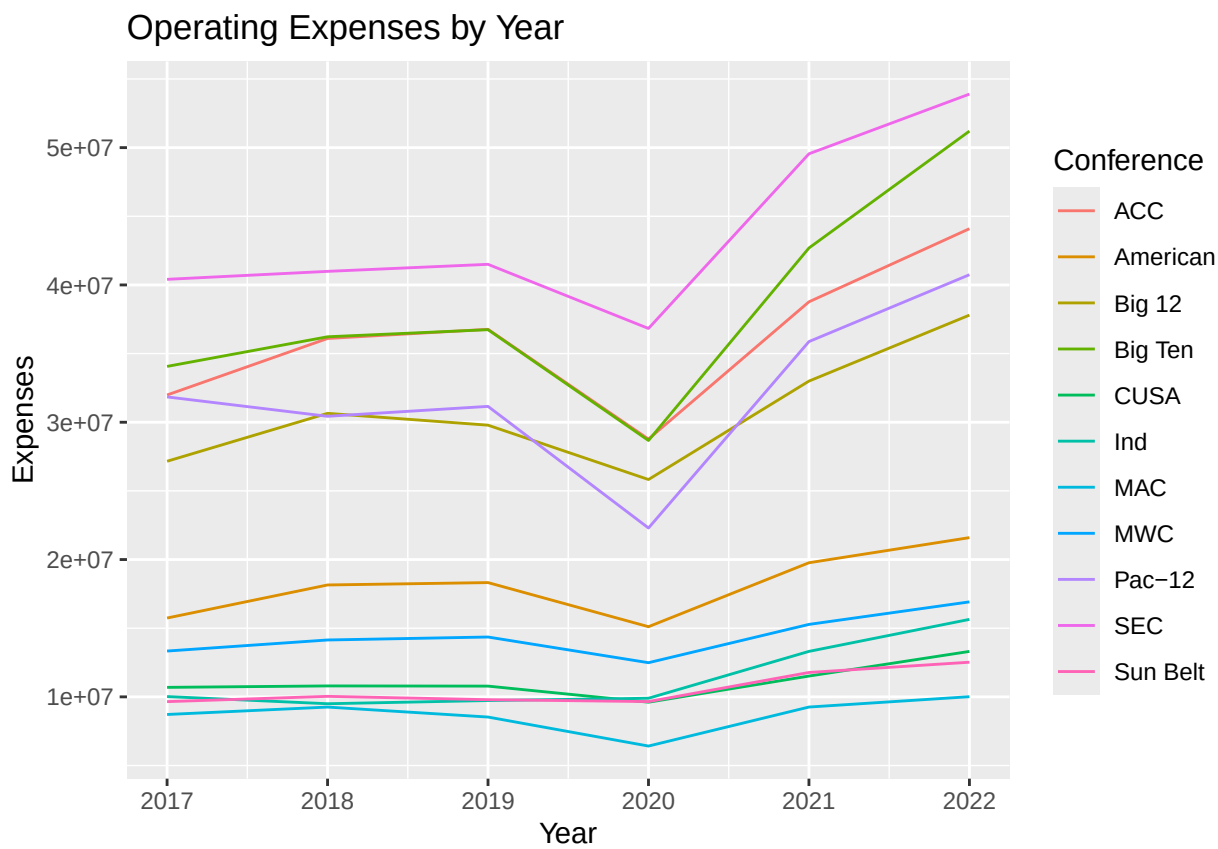
```r
slope19 <-  tidy(mod.fit19)$estimate[2]
slope20 <-  tidy(mod.fit20)$estimate[2]
slope21 <-  tidy(mod.fit21)$estimate[2]
slope22 <-  tidy(mod.fit22)$estimate[2]

slopes <- c(slope17, slope18, slope19, slope20, slope21, slope22)

slopesdf <- data.frame(
  name = c('2017', '2018', '2019', '2020', '2021', '2022'),
  value = slopes
)


ggplot(slopesdf, aes(name, value)) +
  geom_bar(stat='identity', fill = 'steelblue') +
  labs(title = "Slopes by Year", x = "Year", y = "Slope") +
  theme_minimal()
```




```r
conf.by.year <- CFBdata %>%
  group_by(Conf, Year) %>%
  summarise(Avg.Expenses = mean(Football.Total.Operating.Expenses))
```

```
## `summarise()` has grouped output by 'Conf'. You can override using the
## `.groups` argument.
```

```r
ggplot(conf.by.year, aes(Year, Avg.Expenses, color=Conf)) + geom_line() + labs(title = "Operating Expens
```

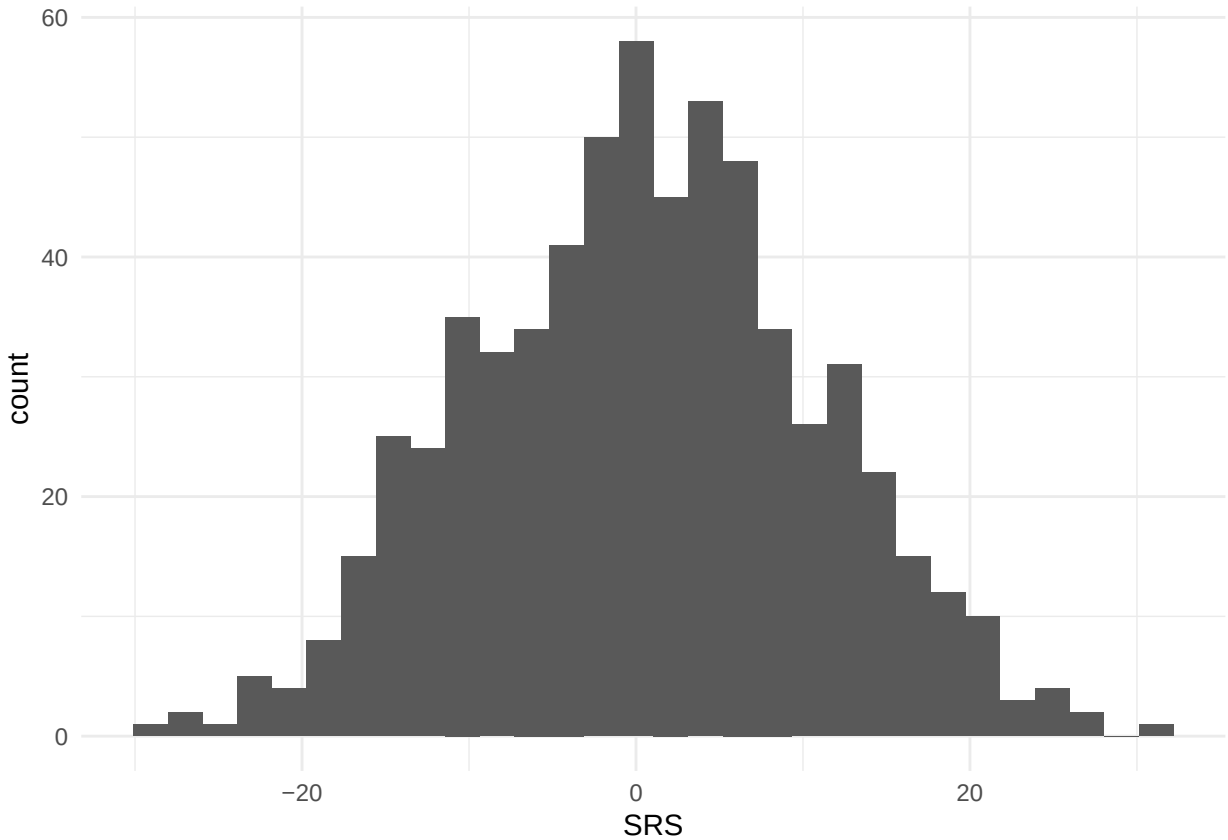



**Strength and Operating Expenses Spread**

```r
ggplot(CFBdata, aes(SRS)) + geom_histogram() + theme_minimal()
```

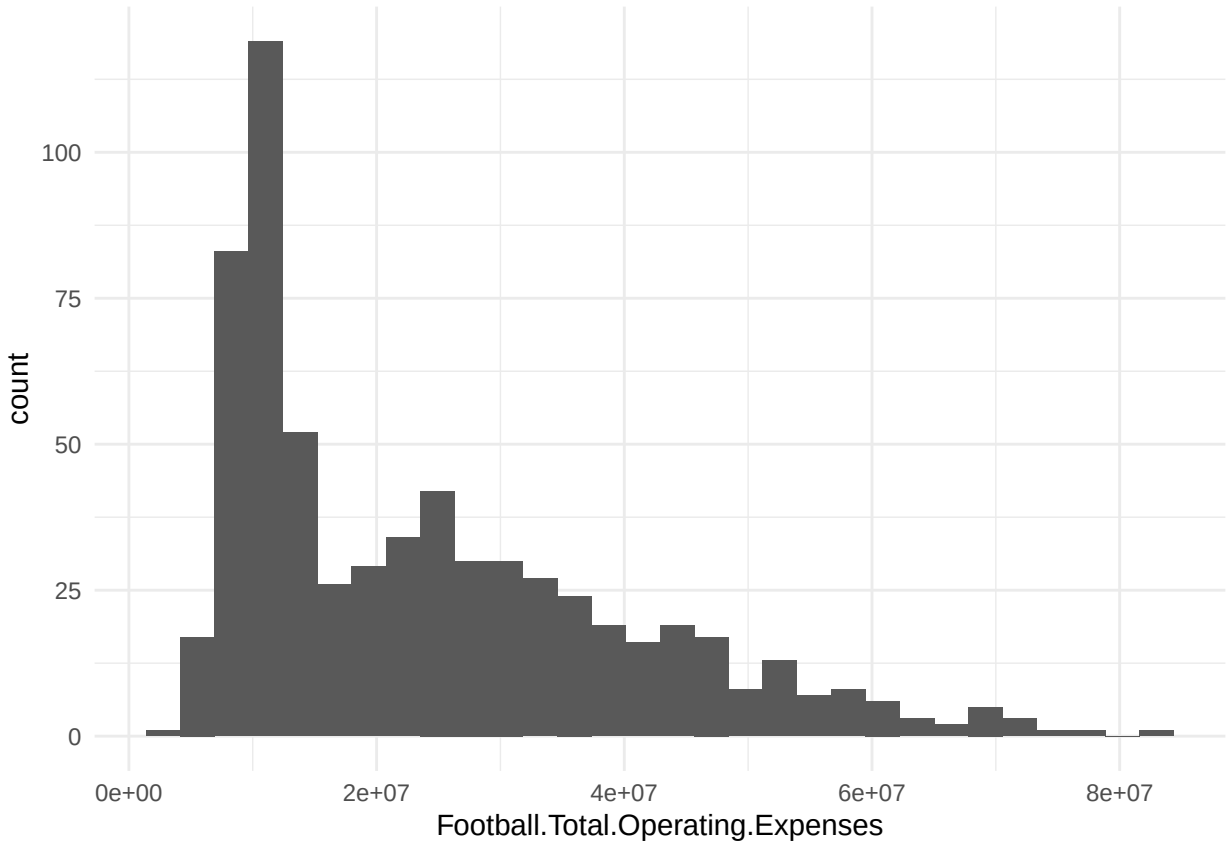
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

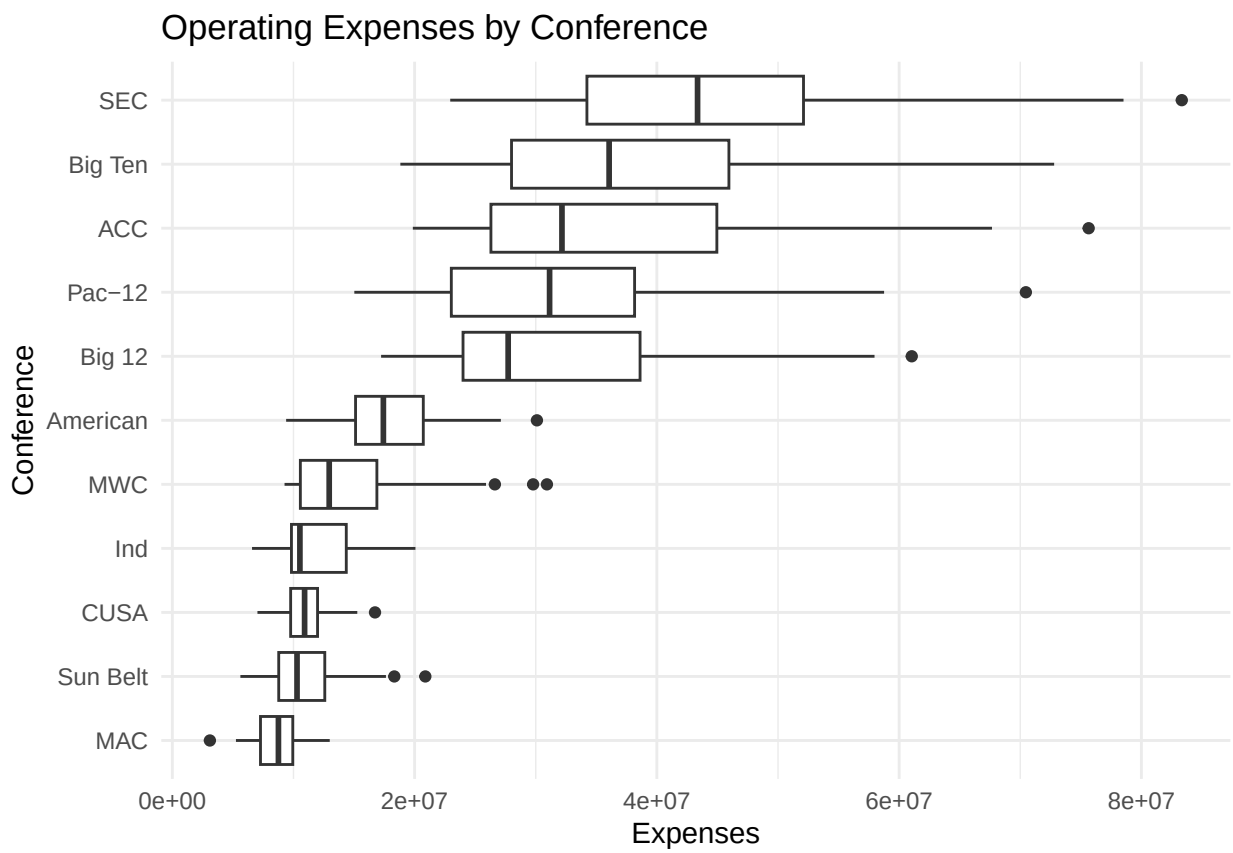



```r
ggplot(CFBdata, aes(Football.Total.Operating.Expenses)) + geom_histogram() + theme_minimal()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
ggplot(CFBdata, aes(Football.Total.Operating.Expenses, reorder(Conf, Football.Total.Operating.Expenses)
```




### Monetary Efficiency

```
Monetary.Efficiency <- CFBdata %>%
  group_by(School) %>%
  summarise(monetary.efficiancy.metric = mean(SRS / Football.Total.Operating.Expenses)) %>%
  arrange(desc(monetary.efficiancy.metric))

Monetary.Efficiency
```

```
## # A tibble: 110 x 2
##    School            monetary.efficiancy.metric
##    <chr>                                  <dbl>
##  1 Appalachian State                0.000000576
##  2 Georgia                          0.000000431
##  3 Oklahoma State                   0.000000398
##  4 Ohio State                       0.000000386
##  5 Cincinnati                       0.000000385
##  6 Boise State                      0.000000378
##  7 UCF                              0.000000370
##  8 Oklahoma                         0.000000354
##  9 Iowa State                       0.000000344
## 10 Clemson                          0.000000343
## # i 100 more rows
```

```r
team_avgs <- CFBdata %>%
  group_by(School) %>%
  summarise(avg.SRS = mean(SRS), avg.expenses = mean(Football.Total.Operating.Expenses))

team_avgs
```

```
## # A tibble: 110 x 3
##    School             avg.SRS avg.expenses
##    <chr>                <dbl>        <dbl>
##  1 Air Force           2.03      10512538.
##  2 Akron             -13.7        8723309.
##  3 Alabama            22.9       68757021.
##  4 Appalachian State   5.87      10691008.
##  5 Arizona            -4.95      27878580.
##  6 Arizona State       2.84      39717305.
##  7 Arkansas            1.30      44444711.
##  8 Arkansas State     -5.67      14770385.
##  9 Army                0.0250    15787870
## 10 Auburn             10.9       52199078.
## # i 100 more rows
```